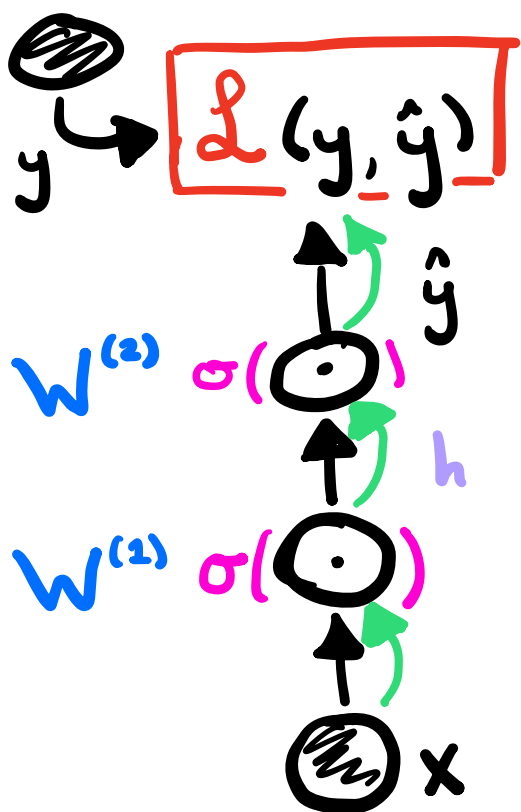


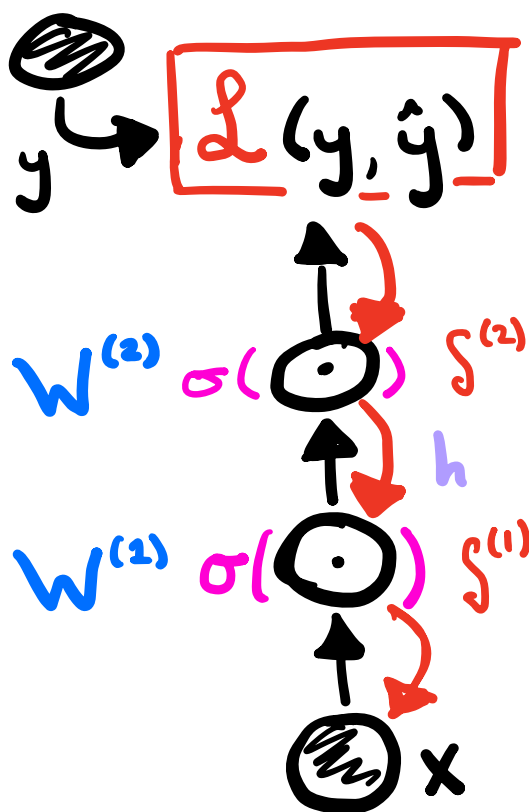
DS 4440

Backpropagation (2)

Last time: Gradients on Computation graphs via backprop.



• forward()



• backward()

For implementation, each node/layer must know how to go forward and backward.

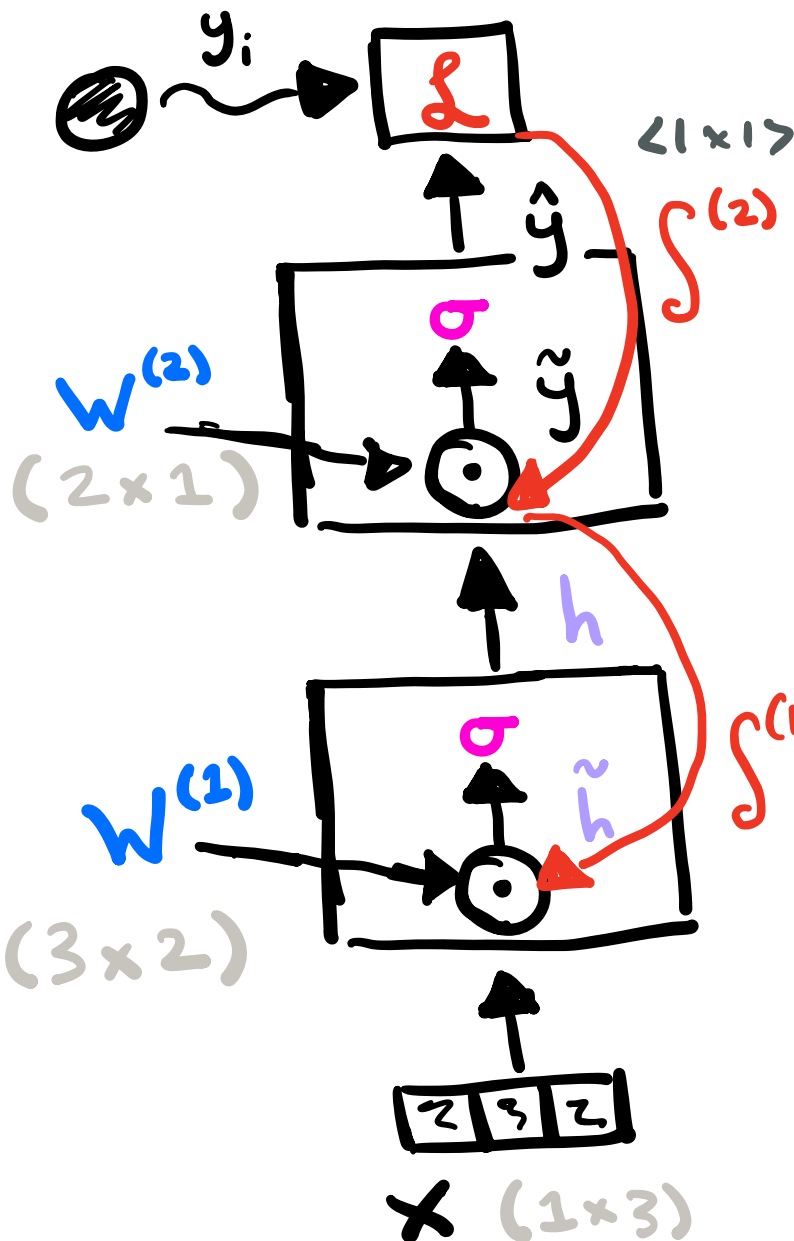
Let's consider an example

$$\mathcal{L} = \text{BCE Loss}(y, \hat{y})$$

$$= -y \lg \hat{y} - (1-y) \lg (1-\hat{y})$$

$$\hat{y} = \sigma(\underbrace{h \cdot W^{(2)}}_{\tilde{y}})$$

$$h = \sigma(\underbrace{x \cdot W^{(1)}}_{\tilde{h}})$$



$$\begin{aligned} \mathcal{J}^{(2)} &= \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \sigma'(\tilde{y}) \\ &= (\hat{y} - y) \end{aligned} \quad \text{See last lecture}$$

$$\mathcal{J}^{(1)} = \underbrace{(\mathcal{J}^{(2)} \cdot W^{(2)T})}_{\text{incoming} \cdot \text{local}} \odot \underbrace{\sigma'(\tilde{h})}_{\frac{\partial h}{\partial \tilde{h}}}$$

$$\nabla_{W^{(2)}} \mathcal{L} = \underbrace{h^T}_{\text{incoming}} \cdot \mathcal{J}^{(2)}$$

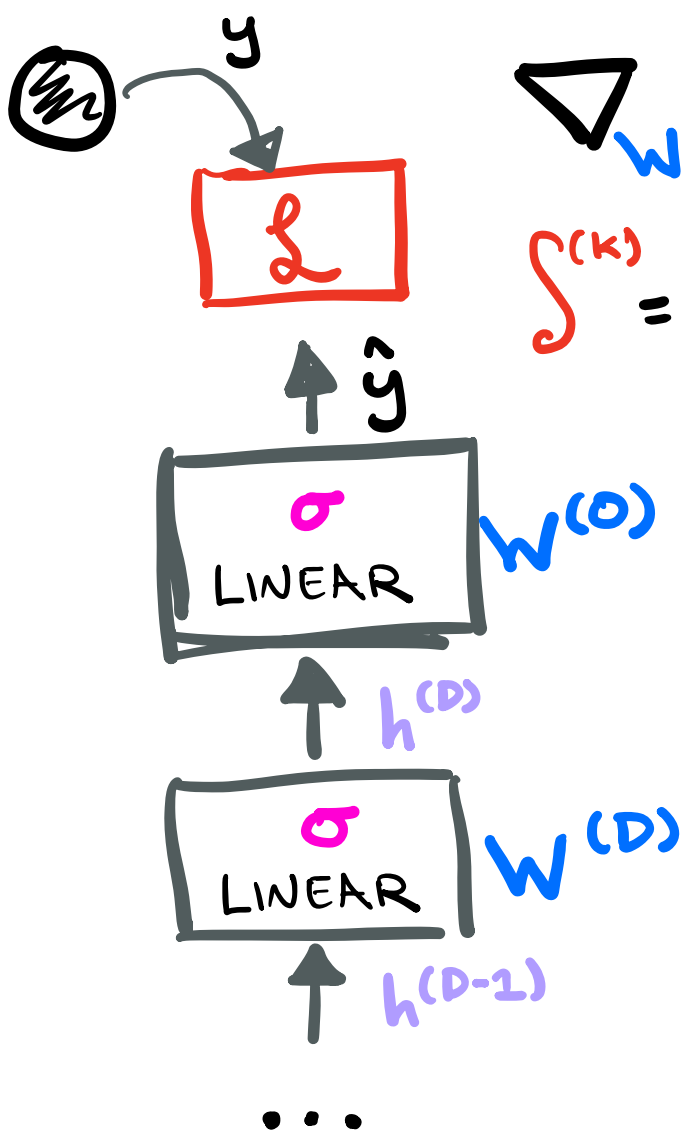
$$\langle 2 \times 1 \rangle \langle 1 \times 1 \rangle \rightarrow \langle 2 \times 1 \rangle \quad \checkmark$$

$$\text{local \cdot incoming} \\ \frac{\partial \tilde{y}}{\partial w^{(1)}}$$

$$\nabla_{w^{(1)}} \mathcal{L} = \underbrace{x^T}_{\text{local \cdot incoming}} \underbrace{\int^{(1)}}_{\frac{\partial \tilde{h}}{\partial w^{(1)}}}$$

$$\langle 3 \times 1 \rangle \langle 1 \times 2 \rangle \rightarrow \langle 3 \times 2 \rangle \checkmark$$

Generalizing: Consider a feed forward network with D layers.

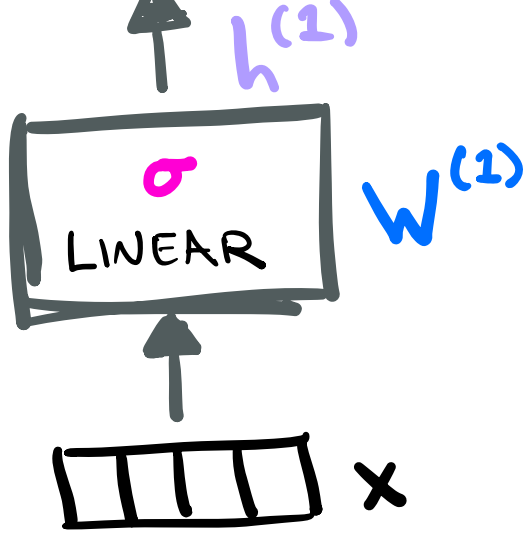


$$\nabla_{w^{(k)}} \mathcal{L} = (h^{(k-1)})^T \int^{(k)}$$

$$\int^{(k)} = (\int^{(k+1)} W^{(k+1)T}) \odot \sigma'(\tilde{h}^{(k)})$$

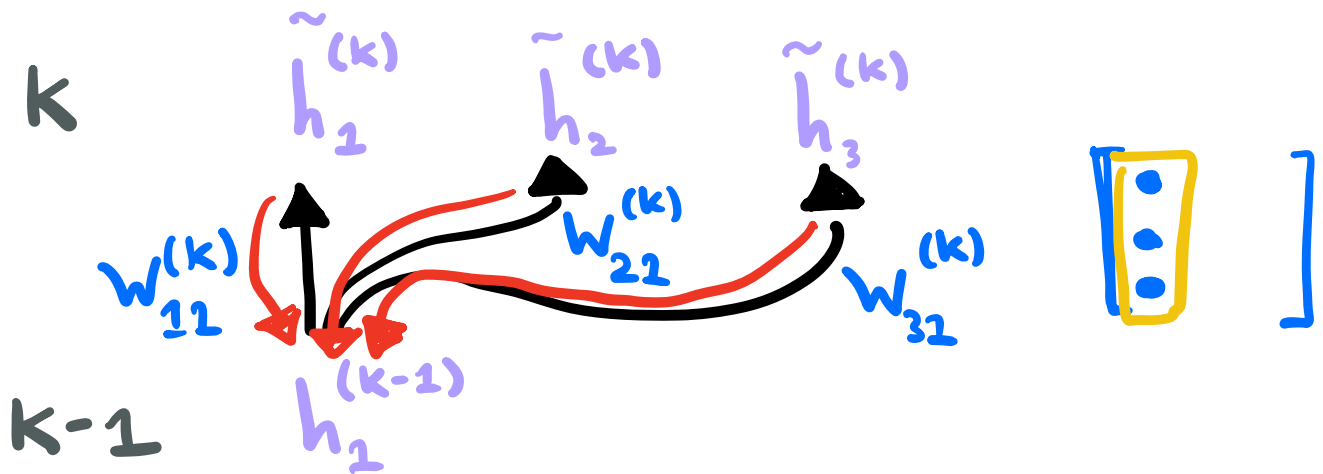
In general we have a Jacobian

$$\frac{\partial \tilde{h}^{(k+1)}}{\partial h^{(k)}} = \begin{bmatrix} \frac{\partial \tilde{h}_1^{(k+1)}}{\partial h_1^{(k)}} & \dots & \frac{\partial \tilde{h}_d^{(k+1)}}{\partial h_1^{(k)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \tilde{h}_1^{(k+1)}}{\partial h_d^{(k)}} & \dots & \frac{\partial \tilde{h}_d^{(k+1)}}{\partial h_d^{(k)}} \end{bmatrix}$$



Will return to This later Today

An intuition for the fully Connected case. Consider just unit $h_1^{(k-1)}$



$$\frac{\partial \tilde{h}_2^{(k)}}{\partial h_1^{(k-1)}} = \frac{\partial W_{21}^{(k)} \tilde{h}_1^{(k-1)}}{\partial h_1^{(k-1)}} = W_{21}^{(k)}$$

So "blame" for error at $\tilde{h}_1^k$ on $h_1^{(k-1)}$ Scales with

1 starts with
Connection weight

Sum over everything it feeds

$$\delta_1^{(k-1)} = \sum_{j=1}^d w_{1j}^{(k)} \cdot \delta_j^{(k)} \cdot \sigma'(\tilde{h}_1^{(k-1)})$$

Influence of $\delta_1^{(k-1)}$ on $\tilde{h}_j^{(k)}$

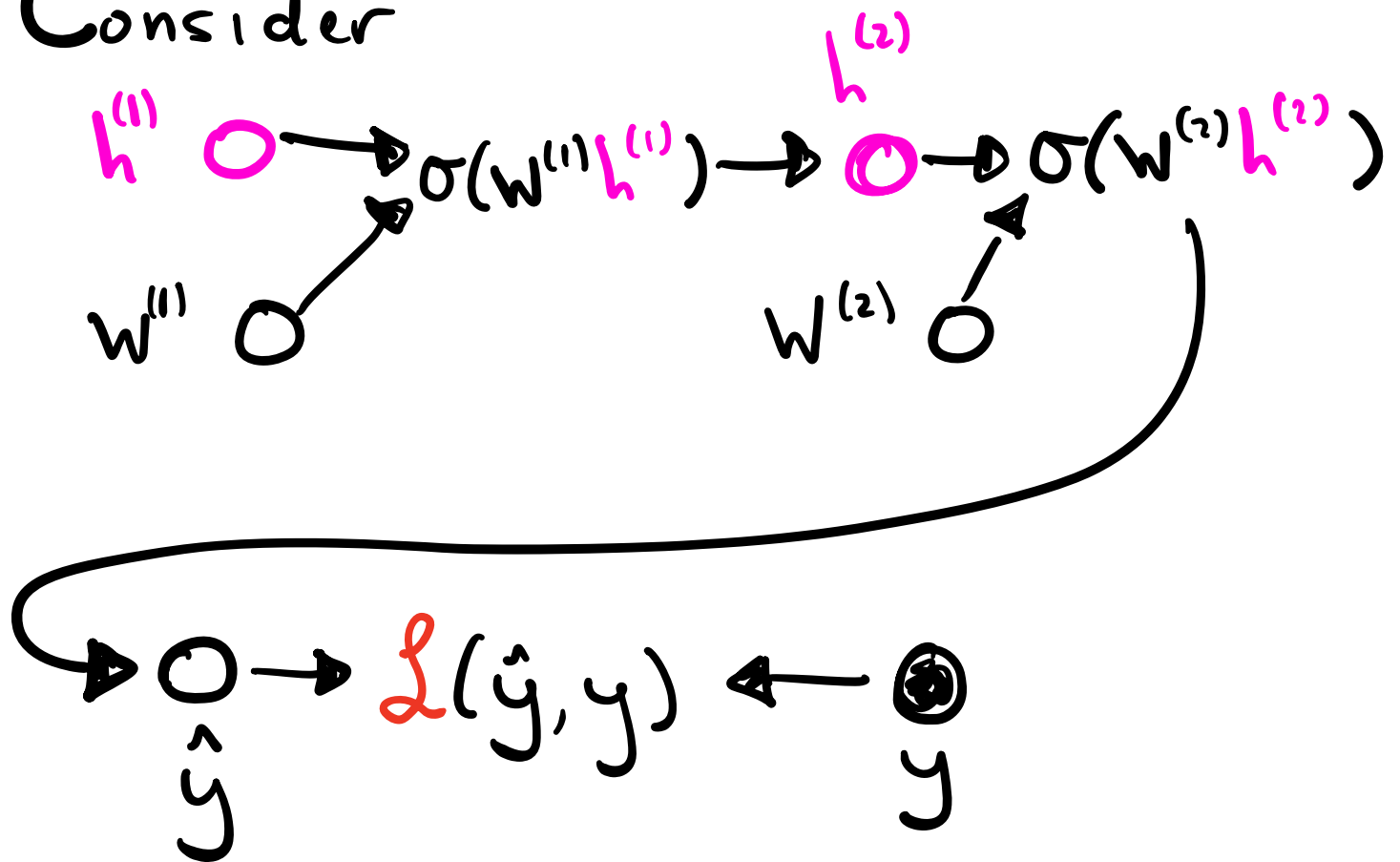
Node error at j

Exercise! Let's implement a wacky custom layer (Colab)

Memory, Backprop & Detach

- What do we need to store for Backprop @ each layer?

• Consider



$$\frac{\partial}{\partial W^{(1)}} =$$

$$\frac{\partial}{\partial \hat{y}} L \cdot \frac{\partial}{\partial h^{(2)}} \hat{y} \cdot \frac{\partial}{\partial h^{(1)}} h^{(2)} \cdot \frac{\partial}{\partial W^{(1)}} h^{(1)}$$

Note That if we freeze lower layers we can discard corresponding activations.

detach()

$$\begin{aligned} y &= x \cdot x \\ z &= y \cdot x \end{aligned} \quad \frac{\partial}{\partial x} z = \frac{\partial}{\partial x} x^3 = 3x^2$$

vs

$$u = y.\text{detach}()$$

erases history of y

$$z = u \cdot x \quad \frac{\partial}{\partial x} z = u$$

When do we actually need the Jacobian?

Layer

J

←

linear;

$$\tilde{h} = h W^{(k)}$$

$$W^{(k)}$$

$$\sum W^{(k)T}$$

elementwise

$$h = \sigma(\tilde{h})$$

diagonal

$$\odot \sigma'(\tilde{h})$$

Coupled outputs

Matrix J

$$\sum J^T$$

Example mean centering

$$y_i = x_i - \frac{1}{d} \sum_j x_j$$

$$J_{ij} = \frac{\partial y_j}{\partial x_i} = 1[i=j] - \frac{1}{d}$$

$$J = I - \frac{1}{d} \mathbf{1} \mathbf{1}^T = \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \quad d=3$$

Not diagonal!

Forward: $x = [1 \ 4 \ 1] \rightarrow y = [-1 \ 2 \ -1]$

Assume incoming $\mathcal{J}^{(y)} = [1 \ 2 \ 6]$

$$\mathcal{J}_i^{(x)} = \sum_j (\mathcal{J}_j^{(y)}) J_{ij}$$

$$= \sum_j (\mathcal{J}_j^{(y)}) (1[i=j] - \frac{1}{d})$$

$$= \sum_j (\delta_j^{(y)}) (1 \mathbb{I}_{i=j})$$

$$- \frac{1}{a} \sum_j (\delta_j^{(y)})$$

$$= \delta_i^{(y)} - \frac{1}{a} \sum_j (\delta_j^{(y)})$$